Main PRD

Sigenic Partner Website 2026 PRD

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Approval

Name	Function	Status	Date
BS	CEO	Not started	Mar 1, 2026
Desmond	COO	Not started	Mar 1, 2026
Viviane	CBDO	Not started	Mar 1, 2026

1. Background & Problem Statement

Sigenic is scaling beyond founder-led support. Currently:

- Customer issues, demo requests, and partner interactions are fragmented across email, chat, and ad hoc channels
- Knowledge is not systematically captured or reused
- Founders remain the bottleneck for issue resolution
- There is no unified interface for partners to:
    - Request support
    - Track issues
    - Access knowledge
    - Evaluate Sigenic's value

If not addressed:

- Support becomes non-scalable
- Knowledge remains tribal
- Customer experience becomes inconsistent
- Growth is constrained by founder bandwidth

2. Product Vision

Create a unified partner-facing portal that serves as: **“The operating system for all partner interactions with Sigenic”**

The portal enables partners to:

- Request and track support
- Access knowledge and best practices
- Submit service and demo requests
- Evaluate ROI through self-service tools

3. Target Users

Primary Users

- Partner engineers and operators
- Customer success and operations teams

Secondary Users

- Internal Sigenic engineers and support staff
- Founder and product team

4. Core Principles

1. Use Zendesk as the support backbone (do not rebuild ticketing)
2. Prioritize self-service over manual support
3. Convert every resolved issue into reusable knowledge
4. Start simple and iterate
5. Portal acts as orchestration layer, not execution layer

5. Solution Architecture (High Level)

Partner Portal (Frontend)

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Integration Layer (APIs / Automation)

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Zendesk (Support Core)

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Execution Layer (Jira / Internal Tools)

↓

Knowledge Base (Notion or Zendesk Guide)

6. Key Features (MVP Scope)

6.1 Support & Ticketing (Core)

Powered by: Zendesk

Features:

- Submit tickets:
  - Bug / Issue
  - General Support
  - Feature Request
- Categorization:
  - Priority
  - Product area
  - Customer impact
- Track status:
  - Open / In Progress / Resolved
- Notifications via email and portal

Design Decision:

- Use Zendesk Web Widget or API integration
- Do not build custom ticketing logic

6.2 Service Request / Intake Layer

Use Cases:

- Demo request
- Feature onboarding
- Data analysis request
- Consulting / advisory support

Implementation:

- Form-based intake (e.g. Typeform or custom UI)
- Routed to Zendesk with tags or internal systems

Purpose:

- Acts as structured sales + operations funnel

6.3 Knowledge Base (Self-Service)

Powered by: Notion or Zendesk Guide

Content:

- FAQs
- Troubleshooting guides
- SOPs
- Known issues
- Best practices

Key Feature:

- Suggested articles during ticket submission

Goal:

- Reduce ticket volume over time

6.4 Success Stories / Case Studies

Purpose:

- Demonstrate value and ROI
- Support sales and adoption

Content:

- Yield improvements
- Excursion prevention
- Before/after comparisons

6.5 Billing & Account (Phase 2)

MVP:

- View invoices (PDF)
- Billing contact details

Future:

- Payment integration
- Usage-based billing

6.6 Authentication & Access Control

MVP:

- Email-based login

Future:

- Role-based access:
    - Admin
    - Engineer
    - Viewer

6.7 Notifications System

Triggers:

- Ticket updates
- Knowledge updates
- Product announcements

Channels:

- Email (MVP)
- In-app (future)

6.8 Score Calculator (Self-Service ROI Engine)

Purpose: Enable partners to simulate the impact of Sigenic solutions before engagement.

Here is the original template: ➕ Risk Score & Yield Loss Simulator

Context Selection:

- Fab / Site
- Department (e.g. Lithography, Etch, CMP)
- Process step
- Machine / tool

Operational Inputs:

- Yield (%)
- Defect rate
- Throughput
- Scrap rate
- Downtime

Business Inputs:

- Cost per wafer
- Volume
- Cost of excursion

System Logic

- Rule-based models (MVP)
- Later upgraded to ML-based models

Output

Estimated Impact:

- Yield improvement (%)
- Scrap reduction (%)
- Downtime reduction (%)

Financial Impact:

- Monthly savings
- Annual savings

Operational Impact:

- Faster detection
- Reduced manual effort

UX:

- Step-by-step wizard
- Dashboard-style results
- Exportable report

Strategic Value:

- Converts portal into revenue-generating tool
- Helps partners justify ROI internally
- Captures valuable baseline data

6.9 Scoring Engine Configuration (Admin System) —

Purpose

Allow Sigenic internal team to:

- Add new machines
- Define process categories
- Configure scoring logic
- Adjust assumptions without code changes

7. Non-Functional Requirements (NFRs)

7.1 User Experience

- Light / Dark mode toggle
- Responsive design (desktop + tablet)
- Intuitive navigation

7.2 Performance

- Page load < 2 seconds
- Ticket submission < 1 second
- Calculator response < 2 seconds

7.3 Security

- HTTPS enforced
- Authentication required
- Secure API integrations

7.4 Scalability

- Support 100–1000 partners initially
- Expand to enterprise scale
- Stateless backend architecture

7.5 Integration Reliability

- Zendesk sync near real-time
- Retry mechanisms for failures

7.6 Knowledge System Quality

- All tickets logged and searchable
- Search latency < 1 second

7.7 Observability

Track

- Ticket volume
- Resolution time
- Calculator usage
- Conversion rates

7.8 Availability

- Target uptime: 99.5%+
- Cloud-hosted infrastructure

7.9 Extensibility

- AI chatbot integration
- Advanced analytics
- Additional modules (billing, dashboards)

8. User Journey

Scenario: Issue Resolution

1. User logs in

2. Searches knowledge base
3. Submits ticket if unresolved
4. Zendesk routes internally
5. Issue resolved
6. Knowledge updated

Scenario: ROI Evaluation

1. User accesses Score Calculator
2. Inputs process and machine details
3. Views estimated savings
4. Clicks “Request Consultation”
5. Ticket created in Zendesk

9. Success Metrics

Operational

- Resolution time ↓
- First response time ↓
- Founder dependency ↓

Product

- Self-service rate ↑
- Knowledge usage ↑

Business

- Customer satisfaction ↑
- Partner retention ↑
- Conversion from calculator → engagement ↑

10. Tech Stack Recommendation

Frontend

- Wix: landing page (out of scope, www.sigenic.com)
- React / Next.js (partner.sigenic.com)

Backend

- Node.js
- Admin Pane: Laravel + Filament

Integrations

- Zendesk API
- Notion API
- Automation tools (Zapier / Make)

11. MVP Scope

Must Have

- Zendesk integration
- Knowledge base
- Service request form
- Score calculator (basic)
- Light/Dark mode

Nice to Have

- Success stories
- Basic analytics

Defer

- Billing system
- Advanced AI
- Complex dashboards

12. Strategic Positioning

This portal is not just a support tool.

It is:

- Support system
- Sales funnel
- Product feedback engine
- Data and AI foundation

Long-term, it evolves into: **Decision Intelligence Platform for semiconductor operations**

Appendix: Admin Dashboard

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Design Principles

It is recommended to use Wix for the public website (Sigenic.com, and use a separate internal admin panel for product logic (partners.sigenic.com).

For that internal admin, Laravel + Filament help with is speed, maintainability, and a good admin experience for non-engineering users. Filament is an open-source Laravel UI framework specifically built to ship admin panels quickly, with built-in support for tables, forms, dashboards, widgets, notifications, and panel configuration. Laravel Nova is also strong, but it is a paid product; Filament is the better default unless you specifically want the official Laravel commercial option.

Why I would lean Laravel for the admin

For the admin use case, the main jobs are:

- Risk Score Configuration:
    - manage machines, manage departments/process labels, edit score assumptions and rules, preview output, review submissions later

This is classic CRUD-heavy internal tooling, and Laravel admin tools are very mature for this. Nova explicitly positions itself as a fast way to manage data, metrics, and custom backend processes, while Filament is built around fast admin delivery with reusable forms and tables.

Alternatively, for Node.js, AdminJS is preferable too. It is a Node-based admin panel option with official docs and is meant for building admin interfaces on top of your backend. Forest Admin is another known option, but it is more of a separate admin product/platform and can add operational complexity; even recent community posts show implementation issues such as auth/CORS friction in some setups.

1. Partner Account Management

Use admin to manage:

- partner companies
- contact persons
- partner status: active, onboarding, inactive
- which sites/fabs belong to which partner
- who can access what
- approval or disabling of accounts

This is useful if one partner has multiple plants, teams, or roles.

2. Access & Permission Control

Beyond Wix member access, admin can control:

- which partner can use score calculator
- which partner can see certain case studies
- which partner can submit service requests
- which partner has billing visibility
- internal staff roles:
    - support admin
    - sales/admin
    - scoring admin
    - super admin

So the admin dashboard becomes your **business permission layer**.

3. Machine / Process / Department Master Data

This is one of the most important admin uses.

Admin should manage:

- departments
- process steps
- machine categories
- machine models

- custom labels
- fab/site mapping
- whether a machine is active/inactive

Without this, every new machine or process update becomes manual engineering work.

4. Scoring & ROI Configuration

This is the brain of your self-service calculator.

Admin can manage:

- score rules
- savings assumptions
- yield improvement assumptions
- downtime assumptions
- cost-per-wafer defaults
- process-specific multipliers
- machine-specific logic
- model versioning

This is probably the highest-value admin module.

5. Request / Intake Management

Even if Zendesk handles ticketing, your dashboard can still help manage:

- demo requests
- onboarding requests
- consultation requests from score calculator
- feature requests
- service upgrade requests

Admin can view:

- request source
- partner name
- request type
- urgency
- owner
- status
- whether it was converted into a Zendesk ticket

This helps your commercial and ops team stay aligned.

6. Ticket Oversight / Escalation View

You may not need to replace Zendesk, but admin can still provide a management layer showing:

- open tickets by partner
- aging tickets
- overdue tickets
- high-priority issues
- repeat issues by machine/process
- escalation flags
- founder-attention-needed items

This is valuable because leaders often want a summary view, not raw Zendesk screens.

7. Success Story / Case Study Management

Admin can manage:

- case studies
- ROI examples
- approved metrics
- logos and testimonials
- visibility rules by partner or industry
- draft vs published content

This lets business teams update proof points without engineering help.

8. Knowledge Base Governance

Even if content lives in Notion or Zendesk Guide, admin can still control:

- which articles are featured
- article categories
- search keywords/tags
- article approval status
- stale content review

- which articles are visible to which partner segment

This makes the portal more curated.

9. Billing / Commercial Visibility

Even if you do not build full payments, admin can manage:

- invoice uploads
- billing contacts
- contract dates
- renewal reminders
- service tier
- package/subscription type
- trial vs paid status

This is useful for account management.

10. Onboarding Workflow

Admin can track:

- new partner created
- kickoff completed
- training completed
- data access granted
- Zendesk access enabled
- calculator enabled
- first case study shared
- adoption milestones

This is very useful if Sigenic wants a repeatable onboarding motion instead of founder-led onboarding.

11. Analytics & Usage Monitoring

Admin dashboard should show:

Partner usage

- who logged in
- which pages are used
- calculator usage
- articles viewed
- requests submitted

Business metrics

- top active partners
- low-engagement partners
- conversion from calculator to consultation
- common machine/process types selected
- support volume by account

This helps both product and GTM.

12. Audit Trail / Change History

Very important for internal trust.

Admin should track:

- who changed score rules
- who added a machine
- who updated assumptions
- who changed permissions
- when invoices or case studies were updated

This matters once more people start operating the system.

13. Announcement / Communication Center

Admin can publish:

- product updates
- maintenance notices
- new feature releases
- training webinars
- support advisories

- onboarding reminders

This can appear on partner dashboards or via email.

14. Form / Workflow Configuration

Instead of hardcoding every request form, admin can manage:

- which request types are enabled
- what fields appear
- mandatory vs optional fields
- routing rules
- which requests go to Zendesk
- which requests go to sales or ops

This makes the portal adaptable.

15. Internal Decision Layer

This is more advanced, but very aligned with Sigenic.

Admin can eventually show:

- which partners have the highest potential savings
- which machine/process combinations create strongest ROI
- where support pain points cluster
- which partners should be targeted for upsell
- which issues should become product roadmap priorities

This turns the admin dashboard into a **management intelligence system**, not just a settings page.